

Remark 05VF on Stacks

ndjiang

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Lemma 1. *Let F be a field. For $1 \leq i \leq r$, let C_i be a smooth, projective, geometrically irreducible curve over F , let $P_i \in C_i(F)$, and let $Q_i \in C_i$ be a closed point with*

$$[\kappa(Q_i) : F] = 2.$$

Put $U_i = C_i \setminus \{Q_i\}$. Then

$$[(P_1, \dots, P_r)] \neq 0 \quad \text{in} \quad \text{CH}_0(U_1 \times_F \cdots \times_F U_r).$$

Proof. Let

$$X = C_1 \times_F \cdots \times_F C_r, \quad U = U_1 \times_F \cdots \times_F U_r,$$

and let $Z = X \setminus U$. The localization sequence for Chow groups gives

$$\text{CH}_0(Z) \longrightarrow \text{CH}_0(X) \longrightarrow \text{CH}_0(U) \longrightarrow 0.$$

Since X is proper over F , there is a degree map

$$\deg : \text{CH}_0(X) \longrightarrow \mathbf{Z}.$$

Every closed point of Z maps to Q_i in at least one factor. Hence its residue field has degree divisible by 2 over F . Therefore every zero-cycle supported on Z has even degree. But

$$\deg[(P_1, \dots, P_r)] = 1.$$

Thus $[(P_1, \dots, P_r)]$ cannot lie in the image of $\text{CH}_0(Z) \rightarrow \text{CH}_0(X)$, so its image in $\text{CH}_0(U)$ is nonzero. $\square$

Theorem 1. *There exist a faithfully flat ring map $A \rightarrow B$ of finite presentation and an A -module M such that $M \otimes_A B$ is free, but M is not locally free.*

Proof. Let

$$k = \mathbf{Q}, \quad k' = \mathbf{Q}(\sqrt{2}).$$

Let $C = \mathbf{P}_k^1$, with affine coordinate t . Let

$$P = \infty \in C(k),$$

and let $Q \subset C$ be the closed point cut out on $\mathbf{A}_k^1$ by

$$t^2 - 2 = 0.$$

Thus $[\kappa(Q) : k] = 2$. Put

$$U = C \setminus \{Q\}, \quad R = \Gamma(U, \mathcal{O}_U),$$

and let

$$\mathcal{L} = \mathcal{O}_U(P), \quad N = \Gamma(U, \mathcal{L}).$$

Then U is affine and N is an invertible R -module.

For $r \geq 1$, set

$$A_r = R^{\otimes_k r}.$$

Thus

$$\mathrm{Spec}(A_r) = U^r.$$

For $1 \leq i \leq r$, define

$$N_{r,i} = R^{\otimes_k(i-1)} \otimes_k N \otimes_k R^{\otimes_k(r-i)}.$$

Equivalently, $N_{r,i}$ is the module of global sections of $\mathrm{pr}_i^* \mathcal{L}$ on U^r .

The transition maps $A_r \rightarrow A_{r+1}$ are given by tensoring with $1 \in R$ in the last factor. Define

$$A = \varinjlim_r A_r.$$

For each $i \geq 1$, define

$$L_i = \varinjlim_{r \geq i} N_{r,i}.$$

Then L_i is the base change of the invertible A_i -module $N_{i,i}$ along $A_i \rightarrow A$. Hence each L_i is an invertible A -module. Now set

$$M = \bigoplus_{i \geq 1} L_i.$$

Thus M is a direct sum of rank-one locally free A -modules.

We first show that M becomes free after a finite etale base change. Put

$$B = A \otimes_k k'.$$

Since k'/k is finite separable, $A \rightarrow B$ is finite etale, hence faithfully flat and of finite presentation.

Let $\alpha = \sqrt{2} \in k'$. On $C_{k'} = \mathbf{P}_{k'}^1$, the rational function $t - \alpha$ has divisor

$$\mathrm{div}(t - \alpha) = [\alpha] - P.$$

The point $[\alpha]$ is one of the two points removed from $U_{k'}$. Hence $t - \alpha$ defines a nowhere-vanishing global section of $\mathcal{O}_{U_{k'}}(P)$. Therefore

$$\mathcal{L}_{k'} \cong \mathcal{O}_{U_{k'}}.$$

Equivalently,

$$N \otimes_R (R \otimes_k k') \cong R \otimes_k k'.$$

It follows that, for every i ,

$$L_i \otimes_A B \cong B.$$

Consequently

$$M \otimes_A B = \bigoplus_{i \geq 1} (L_i \otimes_A B) \cong \bigoplus_{i \geq 1} B,$$

so $M \otimes_A B$ is a free B -module.

It remains to prove that M is not locally free over A . Since U is geometrically integral over k , each A_r is a domain, and the transition maps $A_r \rightarrow A_{r+1}$ are injective. Hence A is a domain.

We show that M_f is not free for every nonzero $f \in A$. Suppose, for contradiction, that M_f is free for some $0 \neq f \in A$. Choose $n \geq 1$ such that $f \in A_n$. Let

$$X_m = U^m = \text{Spec}(A_m)$$

and let

$$\mathcal{E}_m = \bigoplus_{i=1}^m \text{pr}_i^* \mathcal{L}$$

on X_m .

Let $e_1, e_2, \dots$ be a basis of M_f . After enlarging $m \geq n+1$, the elements

$$e_1, \dots, e_{n+1}$$

are represented by sections of $\mathcal{E}_m$ over

$$D(f) \subset X_m.$$

The map $A_m[f^{-1}] \rightarrow A[f^{-1}]$ is faithfully flat. After this faithfully flat base change, the span of $e_1, \dots, e_{n+1}$ is a direct summand of $\mathcal{E}_m$. Therefore the same is true on $D(f)$. Hence the top $n+1$ Chern classes of $\mathcal{E}_m|_{D(f)}$ vanish:

$$c_j(\mathcal{E}_m|_{D(f)}) = 0, \quad j = m, m-1, \dots, m-n. \quad (1)$$

Now let

$$F = k(U^n) = k(t_1, \dots, t_n).$$

Since $f \neq 0$ in A_n , the generic fibre of $D(f) \rightarrow U^n$ is

$$Y = U_F^{m-n}.$$

Restricting (1) with $j = m-n$ to this generic fibre gives

$$0 = c_{m-n} \left(\mathcal{O}_Y^{\oplus n} \oplus \bigoplus_{j=1}^{m-n} \text{pr}_j^* \mathcal{L}_F \right).$$

By the Whitney formula and the identity $c_1(\mathcal{L}_F) = [P]$, this says

$$[(P, \dots, P)] = 0 \quad \text{in} \quad \text{CH}_0(U_F^{m-n}).$$

But $F = k(t_1, \dots, t_n)$ does not contain $\sqrt{2}$, so the base change $Q_F \subset \mathbf{P}_F^1$ is still a closed point of degree 2. Therefore Lemma 1 applies and gives

$$[(P, \dots, P)] \neq 0 \quad \text{in} \quad \text{CH}_0(U_F^{m-n}),$$

a contradiction.

Thus M_f is not free for every nonzero $f \in A$. Since A is a domain, this implies in particular that M is not locally free at the generic point of $\text{Spec}(A)$. Hence M is not locally free. $\square$